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Event Delegation In JavaScript



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What is **Event Delegation**?

Event Delegation is a technique where we attach a **single** event listener to a **parent** element instead of each **child** element. The **event** bubbles up from the child to the parent, and we handle it at the **top** level.

Why use it?

- More **efficient**.
- Works even for **dynamically** added elements.
- **Cleaner** and **scalable** code.



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Project: List Deletion using Event Delegation

Let's build a simple project where:

- You can **add** items to a list.
- You can delete any item by **clicking** a **X** icon using event **delegation**.



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Step 1: HTML (index.html)



```
1 <h2> 🗑 List Deletion with Event Delegation</h2>
2
3 <input type="text" id="itemInput" placeholder="Enter item" />
4 <button onclick="addItem()">Add</button>
5
6 <ul id="itemList"></ul>
```



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Step 2: JavaScript (script.js)



```
1  const itemList = document.getElementById('itemList');
2
3  function addItem() {
4    const input = document.getElementById('itemInput');
5    const text = input.value.trim();
6
7    if (text !== '') {
8      // Create list item
9      const li = document.createElement('li');
10     li.innerHTML = `${text} <span class="delete" style="cursor:pointer; color:red;">X</span>`;
11     itemList.appendChild(li);
12     input.value = '';
13   }
14 }
15
16 // Event delegation: listen on parent, handle delete if clicked on .delete
17 itemList.addEventListener('click', function (e) {
18   if (e.target.classList.contains('delete')) {
19     e.target.parentElement.remove(); // Remove the <li> item
20   }
21 });
```



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🧠 How It Works (Explained Simply):

- The user **adds** an item → we create a new `<li>` with a **X**.
- Instead of **adding** a separate **onclick** for every **X**:
 - We **listen** once on the `ul` (`itemList`).
 - If the **clicked** target has class `delete`, we remove its parent (`<li>`).
- This is **Event Delegation** in action.



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Benefits of Event Delegation in This Project

- We never **rebind** event listeners, even if **100+** items are added.
- Perfect for **dynamic** UIs (e.g., **to-do apps**, **comments**, **chat**).
- Makes code more **maintainable**.



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What You **Learn** from This:

- DOM Manipulation
- Event Handling
- Clean JavaScript practices

Final Thoughts

If you're learning JavaScript, mastering event delegation is a must.

It's one of those core concepts that will make your code clean, scalable, and bug-free.



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